Railway Track Crack Detection Robot

This project presents a smart, low-cost railway track monitoring robot built using an Arduino Uno. The robot follows a simulated railway track using an IR sensor and continuously checks for track gaps or cracks. If a gap is detected, the robot stops automatically and sends an SMS alert along with GPS information using a GSM and GPS module.

Objectives

• Detect cracks or gaps in a railway track.

• Alert maintenance teams using GSM communication.

• Provide location information using GPS.

• Avoid obstacles using an ultrasonic sensor.

• Operate autonomously using battery power.

Hardware Components

• Arduino Uno

• L298N Motor Driver

• IR Line Tracking Sensor

• HC-SR04 Ultrasonic Sensor

• SIM800L GSM Module

• GY-NEO6MV2 GPS Module

• 2 DC Geared Motors

• Battery Pack

• Robot Chassis and Jumper Wires

Working Principle

The IR sensor detects the black line that represents the railway track. The robot moves forward while the line is present. If the line disappears (simulating a crack or missing track), the robot stops and sends an SMS notification with the latest GPS data. The ultrasonic sensor additionally checks for obstacles and prevents collisions.

Key Features

• Autonomous line following

• Railway crack/gap detection

• Ultrasonic obstacle detection

• GSM SMS alerts

• GPS location tracking

• Battery-powered operation

Applications

This system can be used for educational demonstrations, prototype railway inspection systems, and smart infrastructure monitoring projects.

Future Scope

Future improvements include AI-based image processing for real crack recognition, IoT cloud connectivity, multiple IR sensors for improved navigation, and integration with a mobile monitoring application.

Pin Connections

Component	Arduino Pin
IR Sensor OUT	D2
GPS RX / TX	D4 / D5
GSM RX / TX	D6 / D7
L298N IN1	D8
L298N IN2	D9
L298N IN3	D10
L298N IN4	D11
Ultrasonic TRIG	D12
Ultrasonic ECHO	D13

Conclusion

The Railway Track Crack Detection Robot demonstrates how embedded systems, sensors, GPS, and GSM technologies can be combined to create an intelligent railway safety solution. The project provides a practical and scalable approach for improving railway inspection and reducing accident risks.